

Rainfall Estimation

This note summarizes results for a collection of 100 patches. All aggregate figures and tables that follow should therefore be read as patch-level summaries over this hundred-patch evaluation set, rather than as statistics from a single patch or from the full archive of available patches.

Relative Rainy-Pixel Distance Profile

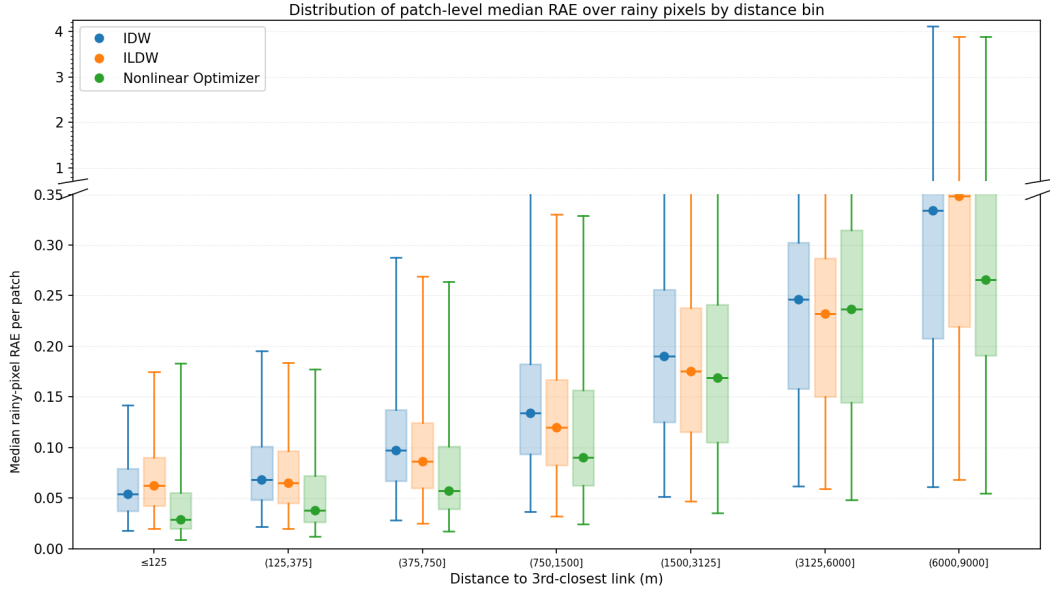


Figure 1: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6), and relative absolute error for a rainy pixel p is defined as $|GT(p) - PRED(p)|/GT(p)$.

This plot summarizes how solver performance changes as pixels get farther from the third-nearest link. Lower values indicate lower rainy-pixel relative absolute error in that distance bin.

Rainy-pixel Count Table for Distance to the 3rd Closest Link

The table below gives the average number of rainy pixels per patch and the corresponding standard deviation for each distance bin to the 3rd closest link. These are the same patch-level counts summarized in the tick labels of the rainy-pixel distance-profile plots.

Distance to third closest link (m)	Mean rainy-pixel count per patch (% of total)	SD
[0, 125]	4267.0 (1.79%)	960.4
(125, 375]	16737.7 (7.01%)	3803.2
(375, 750]	26809.9 (11.22%)	6707.0
(750, 1500]	51988.5 (21.76%)	14 468.0
(1500, 3125]	88839.4 (37.18%)	27 531.1
(3125, 6000]	45929.4 (19.22%)	15 679.4
(6000, 9000]	3835.8 (1.61%)	2725.7
(9000, ∞)	514.4 (0.22%)	966.9
Total	238922.1 (100%)	

Table 1: Distribution of the number of rainy pixels per distance bin. In each patch, rainy pixels are classified into bins according to their distance to the 3rd-closest link. Formally, a pixel p belongs to bin $(d_1, d_2]$ if the disc of radius d_2 centered at p intersects at least 3 links and the disc of radius d_1 centered at p intersects at most 2 links. Let $|\text{bin}_i(P)|$ denote the number of rainy pixels in patch P that belong to bin_i . The table reports, for each bin, the mean and standard deviation of $|\text{bin}_i(P)|$ across patches, with the percentage of the total mean rainy-pixel count shown in parentheses next to the mean. The total is the sum of the per-bin mean counts. The last bin, $(9000, \infty)$ is not shown in Figure 1 because it is only sparsely populated.

Confusion Map

IDW			ILDW			Nonlinear Optimizer		
GT	Prediction		GT	Prediction		GT	Prediction	
	Wet	Dry		Wet	Dry		Wet	Dry
Wet	0.893 ± 0.013	0.002 ± 0.000	Wet	0.892 ± 0.013	0.003 ± 0.000	Wet	0.890 ± 0.013	0.005 ± 0.001
Dry	0.040 ± 0.004	0.065 ± 0.010	Dry	0.037 ± 0.004	0.068 ± 0.010	Dry	0.030 ± 0.004	0.075 ± 0.010

Table 2: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM). These are therefore fractions of all pixels in a patch, not row-normalized conditional rates.

Patch Geometry Overview

Approx. patch size (km)	# links	1-multiplicity	2-multiplicity	3-multiplicity
50×50	939	99	840	0
62×60	1254	128	1126	0
72×70	1507	157	1350	0

Table 3: Representative link multiplicities for three HundredPatches cases whose dimensions are close to 50×50 , 60×60 , and 70×70 km. Here, multiplicity is defined by grouping links whose corresponding endpoints match within 1 m in Euclidean distance, allowing endpoint order reversal. A 1-multiplicity link appears alone, a 2-multiplicity link belongs to an exactly duplicated pair, and a 3-multiplicity link belongs to a triplet of coincident links. The entries in the table count links, not multiplicity groups.

J_{atten} Ratios

Solver	Mean J_{atten}	SD
IDW	0.01522	0.03808
ILDW	0.00289	0.00757
Nonlinear Optimizer	0.00020	0.00103

Table 4: Patch-level summary of J_{atten} across the 100 evaluated patches. For a given patch, J_{atten} is the attenuation-fit term $J_{\text{atten}} = \frac{1}{N_{\text{valid links}}} \sum_{k \in \text{valid links}} \frac{(\hat{A}_k - A_k^{\text{obs}})^2}{L_k^{\text{km}}}$, where $\hat{A}_k$ is the attenuation implied by the estimated rainfall field on link k , A_k^{obs} is the observed attenuation, and L_k^{km} is the link length in kilometers. The table reports the mean and standard deviation of this per-patch quantity for each solver. Lower values indicate better agreement with the observed link attenuations.

Ratio of solver means	Value
$\text{Mean}(J_{\text{atten}}(\text{IDW})) / \text{Mean}(J_{\text{atten}}(\text{ILDW}))$	5.28
$\text{Mean}(J_{\text{atten}}(\text{ILDW})) / \text{Mean}(J_{\text{atten}}(\text{Nonlinear Optimizer}))$	14.15

Table 5: Ratios formed after first averaging J_{atten} over patches within each solver. These values therefore compare solver-level means, in contrast to the table below, which averages the per-patch solver-to-solver ratios.

Ratio	Mean over patches
IDW / ILDW	6.00
ILDW / Nonlinear Optimizer	437.71

Table 6: Patch-level ratios of J_{atten} across the 100 evaluated patches. For a given patch, J_{atten} is the attenuation-fit term $J_{\text{atten}} = \frac{1}{N_{\text{valid links}}} \sum_{k \in \text{valid links}} \frac{(\hat{A}_k - A_k^{\text{obs}})^2}{L_k^{\text{km}}}$, where $\hat{A}_k$ is the attenuation implied by the estimated rainfall field on link k , A_k^{obs} is the observed attenuation, and L_k^{km} is the link length in kilometers. For each patch, we compute the solver-to-solver ratio shown in the left column, and the table reports the average of those per-patch ratios across the full hundred-patch set. Because lower J_{atten} is better, a large value of, for example, ILDW / Nonlinear Optimizer means that the nonlinear optimizer achieves much smaller attenuation-fit error than ILDW on a typical patch.

Absolute Attenuation Error per km

Solver	Mean	SD
IDW	0.05034	0.03413
ILDW	0.01356	0.00948
Nonlinear Optimizer	0.00230	0.00437

Table 7: Patch-level summary of the mean absolute attenuation error per km. For a given patch, we compute $\frac{1}{N_{\text{valid links}}} \sum_{k \in \text{valid links}} \frac{|\hat{A}_k - A_k^{\text{obs}}|}{L_k^{\text{km}}}$, where $\hat{A}_k$ is the attenuation implied by the estimated rainfall field on link k , A_k^{obs} is the observed attenuation, and L_k^{km} is the link length in kilometers. Thus, each valid link contributes equally after its own absolute attenuation error is normalized by link length. The table reports the mean and standard deviation of this per-patch quantity across the 100 evaluated patches for each solver. Lower values indicate better agreement with the observed link attenuations.

Solver	Mean	SD
IDW	0.05512	0.04126
ILDW	0.00877	0.00700
Nonlinear Optimizer	0.00211	0.00365

Table 8: Patch-level summary of the globally length-normalized absolute attenuation error. For a given patch, we compute $\frac{\sum_{k \in \text{valid links}} |\hat{A}_k - A_k^{\text{obs}}|}{\sum_{k \in \text{valid links}} L_k^{\text{km}}}$ using the same notation as above. Unlike the previous table, this metric first sums the absolute attenuation error over all links in the patch and then divides by the total link length, so longer links receive more weight. The table reports the mean and standard deviation of this per-patch quantity across the 100 evaluated patches for each solver. Lower values again indicate better agreement with the observed link attenuations.